

BMC351: PYTHON PROGRAMMING LAB ASSIGNMENTS LIST

Write Python programs for the following:

1. Swap two numbers without using a third variable and also without using arithmetic operators
2. Print all the data types in python along with their sizes
3. Implement the working of all the operators in python using a menu based program
4. Check whether a year is a leap year or not
5. Print Prime numbers in a given range
6. Print Armstrong numbers in a given range
7. Determine student grade based on marks using if-elif-else
8. Verify whether a character is a vowel or consonant
9. Check if a triangle is isosceles, equilateral, scalene, right-angled or Invalid
10. Generate multiplication table for a number using loops
11. Count the number of alphabets and digits in the string 'Nagpur-440010'
12. Find the reversed number of a 5-digit input and check if it's equal to the original
13. Remove all the multi-spaces to single space and then reverse a string
14. Count vowels, consonants, digits, and special characters in a string
15. Check whether a string is a palindrome
16. Convert string to uppercase, lowercase, and title case
17. Print different alignments of a string
18. Replace all occurrences of a word in a string with another word
19. Find largest and smallest element in a list
20. Merge two lists and remove duplicates
21. Count frequency of each element in a tuple
22. Dictionary of students with marks: find highest and lowest
23. Create a Python function to merge two dictionaries and return sorted keys
24. Create two lists: one containing positive numbers and one containing negative numbers from an original list
25. Convert a list of 5 strings to uppercase
26. Find the number of negative numbers in a list of integers
27. Create a list of first characters from a list of words
28. Find the number of days between two dates represented as tuples (d,m,y)
29. Sort a list of (item, price) tuples in descending order by price
30. Implement all types of slicing in string, list, tuple
31. Update a student marks dictionary to replace subject marks with total and average, and report the topper
32. Implement working of all types of arguments to a function
33. Using iteration and recursion implement the following: factorial, Fibonacci, GCD, Sum of first n natural numbers
34. Using random module, simulate 2 dice rolls until a sum of 10 or more is achieved
35. Write a Python program to find GCD and LCM of two numbers using functions

36. Define `count_lower_upper()` function to calculate uppercase/lowercase counts in a string and return as a dictionary
37. Read a text file and count the number of lines, words, and characters
38. Read a file and display its contents with line numbers before each line
39. Append the contents of one file at the end of another
40. Read student records (name, age) from a file and display them sorted by name
41. Merge lines alternately from two files into a new file
42. Find the sum of two numbers using command line arguments
43. Use command-line arguments to search for a word in a file and replace it with a specified word
44. Emulate type and copy command of DOS using command line arguments
45. Write a program to handle multiple exceptions (e.g., division by zero, invalid input)
46. Receive 5 integers and store them and their cubes in a dictionary; raise user-defined exceptions `NumberTooSmall` (if <3) and `NumberTooBig` (if >20)
47. Implement transpose, addition, subtraction, multiplication of 2 matrices using numpy
48. Implement basic operations on Series and DataFrame using pandas
49. Implement search, match, sub functions of re module
50. Find the number of days between two dates using datetime module